

Design for impactful communication

Presentation design principles

Presentation design principles

Creating effective slides in a presentation involves **more than just visually appealing design**. It also **requires a clear and logical flow of information**.

An effective presentation is a **harmonious combination of your narrative, visuals, and the slide design**, all working together to convey our message persuasively and memorably.

To accomplish all of this, we need to ensure the **clarity, visual appeal, consistency**, and many other principles of the presentation slides we create.

The key principles

Audience engagement



A **clear and logical flow** of information **allows viewers to stay engaged** with the content and **understand the information** being presented.

Slides should convey **information clearly and concisely**. **Avoid clutter**, and use a **minimal amount of text**. Each slide should have a **singular, focused message or idea**.



Clarity

Visual appeal



Visually appealing design elements, used thoughtfully, can **reinforce our message and make the content more engaging**.

Consistency helps **create a cohesive and professional look**. Use the **same font, colours, and formatting** for headings, bullet points, and other text elements.



Consistency

Storytelling



Organise slides in a **logical sequence that tells a story**. Each slide should build on the previous one, **creating a narrative flow** that helps the audience follow our presentation more easily.

How do we implement the principles?

Implementing the key principles for **impactful communication** requires a combination of what we've learned about:

Storytelling



Visuals



Data



Narrative

Design



Colour



Proximity



White space



Alignment



Contrast



Hierarchy

Accessibility



Purpose



Content



Structure



Formatting

How do we implement the principles?

Layout and composition



Text and typography



Colour, contrast, and harmony



Images and graphics



Transitions and animations



Layout and composition

Simplicity: Keep slides uncluttered. Avoid overcrowding with excessive text or graphics. A clean and minimalistic design ensures that our audience can focus on the key message.

What is Scaling?

Scaling is the process of **adding** or **removing compute resources** based on workload demands.

To fully understand how scaling works, it is important to be familiar with the **terminology** related to scaling.

An instance is a **virtual server** in a cloud environment.

Up: **Increases the size** and capacity of compute resources.
Down: **Decreases the size** and capacity of compute resources.
Out: **Adds more compute resources of the same size.**

A Load Balancer efficiently **distributes network traffic** across a group of instances.

The process of **automatically** increasing or decreasing compute resources.

Increasing the **size** and **computing power** of an instance. The number of instances remain the same.

Increasing the **number of instances**. The size and computing power of the instances remain unchanged.

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Automatic scaling and monitoring

What is scaling?

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Instance An instance is a virtual server in a cloud environment.	Auto-scaling The process of automatically increasing or decreasing compute resources .	Vertical scaling Increasing the size and computing power of an instance. The number of instances remain the same.
Horizontal scaling Increasing the number of instances. The size and computing power of the instances remain unchanged.	Load balancer A load balancer efficiently distributes network traffic across a group of instances.	Scaling up, down, and out Up increases and down decreases the size and capacity of compute resources. Out adds more compute resources of the same size .

18

These two examples include the **exact same information**, however, there's a **trade-off**: simplicity makes it easier to follow but fewer visual representations of a concept might make it harder to interpret at first.

Layout and composition

Balance: Distribute elements evenly across the slide. Achieve balance by arranging content in a visually appealing way, avoiding heavy concentration on one side of the slide.

The AWS Well-Architected Tool (WAT)

The AWS WAT is a cloud service that provides a consistent process for measuring your architecture using AWS best practices

Architectural Guidance

Enable Consistent Governance

Continuously Improve Workloads

Access the AWS Solution Architect Knowledge Base

Amazon WAT

Review Cloud Resources

Understand Workload Risks

AWS WAT Use Cases:

- Assist with the **documentation** of specific **architectural decisions** made.
- Provide **recommendations** for **improving** your **workload-based** on **best practices**.
- Guide** you in making your **workloads** **more reliable, secure, efficient, and cost-effective**

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Architectural guidance 01.

Continuously improve workloads 02.

Access the AWS solution architect knowledge base 03.

Enable consistent governance 04.

Review cloud resources 05.

Understand workload risks 06.

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25

Balance can also be achieved by using **consistent font and element sizes**.

Layout and composition

White space: Make effective use of white space (empty space) around elements on the slide. It provides visual “breathing room” and helps guide the viewer's eyes to the most important content.

Amazon EC2 Auto Scaling

AWS has several options for auto scaling. When you want to **focus** on **managing** your **EC2 instances**, [Amazon EC2 Auto Scaling](#) will be your option.

Benefits	EC2 Auto Scaling Management Options		
- Increases and maintains your application availability. - Improves fault tolerance. - Automatically adds or removes instances based on predefined conditions. - Different management options. - Lowers cost. - No charge for using the EC2 auto scaling feature.	Fleet Management An EC2 Fleet is a group of On-Demand or Spot Instances. Fleet Management maintains the health of your fleet. Detects unhealthy instances. Terminates impaired instances and replaces them. Launch new instances across Availability Zones in a balanced way.	Dynamic Scaling Dynamic scaling responds to changing demand and automatically increases or decreases the capacity. If CPU spikes above or below a threshold you have set (e.g. 80%), a new instance will get added or removed dynamically.	Predictive Scaling A machine learning algorithm predicts future traffic and regular spikes by detecting daily and weekly patterns. The right number of instances are then provisioned, in advance, accordingly.

Read more [here](#) on the Auto Scaling policies and groups regarding specifications on the number of instances.

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Automatic scaling and monitoring

Amazon EC2 Auto Scaling

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EC2 Auto Scaling management options

Fleet Management	Dynamic Scaling	Predictive scaling
An EC2 fleet is a group of on-demand or spot instances. Fleet Management maintains the health of the fleet: Detects unhealthy instances. Terminates impaired instances and replaces them. Launch new instances across Availability Zones in a balanced way.	Dynamic scaling responds to changing demand and automatically increases or decreases the capacity. If CPU spikes above or below a threshold we have set (e.g. 80%), a new instance will get added or removed dynamically.	A machine learning algorithm predicts future traffic and regular spikes by detecting daily and weekly patterns. The right number of instances are then provisioned in advance.

21

We often need to **move some of the information to a next slide**, **reduce the amount of text and/or graphics**, etc. to achieve effective “breathing room” on a slide.

Layout and composition

Alignment: Ensure that text and graphics are aligned properly. Consider using grids and guidelines to assist in aligning elements precisely. This can help us create a structured and organised layout.

The benefits of using Amazon Route 53



Highly Available, Reliable & Flexible

- Amazon Route 53 Traffic Flow re-routes traffic if an application endpoint is unavailable.
- Multiple traffic policies can be defined and edited in the Route 53 console, AWS SDKs or the Route 53 API.

Simple & Fast

- Configure DNS settings in the AWS Management Console or the AWS API.
- Use the self-service sign-up.
- Users will be automatically routed to the best location based on network conditions.

Cost-effective

- Only pay for resources being used:
 - Number of queries answered per domain.
 - Hosted zones.
 - Traffic flows and health checks.
- No minimum usage commitments or up-front payments.



Secure

- Grant unique credentials and permissions to users by integrating with AWS IAM.



Scalable

- Automatic scaling to handle large query volumes.

Use with other services

- Designed to work with other AWS features and offerings.
- Provides a hybrid cloud solution by connecting with your on-premise networks.

Read more [here](#) on Amazon Route 53 features.

EXPLORE DIGITAL SKILLS

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The example on the right has the **guides option activated**. This allows us to see whether or not our **elements are aligned properly**.

Text and typography

Hierarchy: Use a clear visual hierarchy to highlight important points. Larger fonts, bolder text, or different colours can be employed to distinguish main ideas from supporting information.

Authorisation

Authorisation is the process of determining what permissions a user or service should be granted. By default IAM users do not have permission to access anything. You are required to **explicitly grant permissions** by creating a policy and attaching it to the user

Principle of least privilege (PoLP)

It is the principle of applying the **minimal set of permissions needed to accomplish a specific task**. Determine what users need to be able to do, and then create policies so that they can perform their duties.

In the AWS cloud, most resources are sealed shut when they are created, and you provide access to them using a policy attached to a user, a group, or a role.



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AWS cloud security

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In the AWS cloud, **most resources are sealed when they are created**, and we can provide access to them using a policy attached to a user, a group, or a role.

We can use font **size**, font **colour**, **coloured text boxes**, and **other forms of formatting** to ensure that the more important concepts are highlighted.

Text and typography

Alignment: Ensure that text and graphics are aligned properly. Left alignment is often preferred for text, as it makes content easier to read.

Working to ensure compliance

AWS continuously interacts with certifying bodies and independent auditors to provide customers with detailed information about the policies, processes, and controls that are established and operated by AWS.

Certifications and attestations

Certifications and attestations are assessed by a third-party, independent auditor.

Examples:

- ISO 27001, 27017, 27018
- ISO/IEC 9000

Laws, regulations, and privacy

AWS provides security features and legal agreements to support compliance.

Examples:

- EU General Data Protection Regulation (GDPR).
- Health Insurance Portability and Accountability Act (HIPAA)

Alignments and frameworks

AWS aligns on industry or function-specific security or compliance requirements.

Examples:

- Center for Internet Security
- EU-US Privacy Shield certified.

EXPLORE DIGITAL SKILLS

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Although **left alignment is most preferred** for text, centre and right alignment can be used effectively in very specific situations.

Text and typography

Typography: Choose legible fonts for your text. Sans-serif fonts like Arial or Helvetica are often easier to read on screens. Maintain a reasonable font size, typically around 24pt or higher for presentations, to ensure readability.

Table Relationships

Relational databases are collections of tables that are connected together. Each table represents an entity that has a collection of attributes (i.e. table columns). Tables in a database are connected to each other by means of relationships which are specified according to how different entities interact with each other. There are 4 main types of table relationships:

Relationship	ER diagram Symbol	Description
One to One		An instance in one table relates to a single instance in another table, e.g. a customer in the customers table can only have one address in the addresses table.
One to Many or Many to One		An instance in one table can correspond to multiple instances in another table, e.g. each customer from the customers table can have multiple invoices from the invoices table. For the "Many to One" relationship, the reverse is true.
Many to Many		Multiple instances of one table can correspond to multiple instances of another table, e.g. one invoice can contain multiple items and each item can be in multiple invoices.
Self Referencing Relationship	* depends on nature of relationship	An instance in one table can correspond to another instance in the same table, i.e. through a different attribute.

Creating a relational database

Table relationships

Relational databases are **collections of tables that are connected**. Each table represents an **entity** that has a **collection of attributes** (i.e., columns). Tables in a database are **connected to each other by means of relationships** which are specified according to how different entities interact with each other.

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Our choice of **font colour** can also contribute to readability.

Colour, contrast, and harmony

Colour scheme: Select a colour scheme that is visually appealing and conveys the message effectively. We need to be mindful of colour accessibility, ensuring that our colour choices are suitable for all audiences.

Why should I use Amazon VPC?

Amazon Virtual Private Cloud (VPC) allows you to create a virtual isolated network in the cloud. In this network space you can define how your network and resources are exposed to the internet. You will also have complete control over IP address ranges, subnets, [route tables](#) and network gateways.

Benefits of using Amazon VPC



Secure

- AWS VPC security groups and network ACLs act as firewalls.
- You can configure rules for inbound and outbound traffic.
- You can enable access control with an [VPC endpoint](#).



Simple

- An Amazon VPC can easily be created in the AWS Management Console or [AWS CLI](#).
- The Amazon VPC console wizard provides multiple options to create a VPC, you can select one suitable to your needs and customize it.
- All components of the VPC are automatically created by the VPC wizard.



Customisable

- Configure the Amazon VPC to fit your specific needs.
- Choose your IP Address range and create your own subnet.
- Configure route tables to available gateways.

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Networking and content delivery

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Colours that are **too similar can make it difficult to differentiate** between key concepts and messages.

Colour, contrast, and harmony

Using contrast: Use contrasting colours for text and backgrounds to enhance interpretability and readability.

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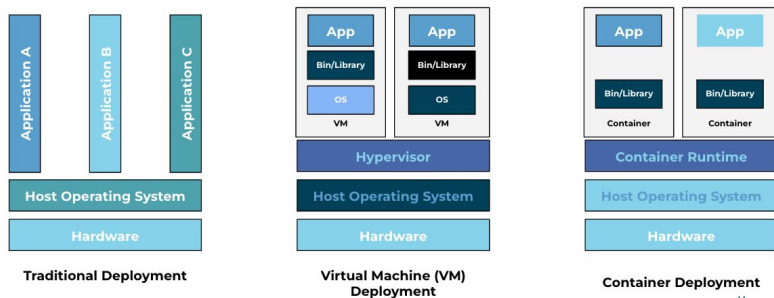
The use of contrasting colours is extremely important in **ensuring accessibility**.

Colour, contrast, and harmony

Intentional colour use: Use colour to evoke emotion, create connection, and tell stories.

Containers vs. VMs

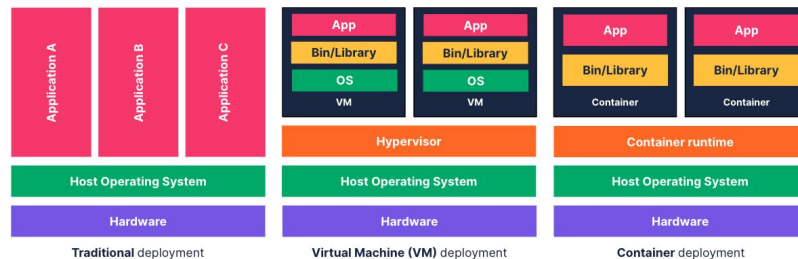
While a virtual machine virtualises an entire computer (OS, hardware, networking, etc), a container virtualises an entire operating system. The significant difference between containers and VMs is that containers share the host's operating system with other containers.



AWS compute services

Containers versus VMs

While a **virtual machine virtualises an entire computer** (OS, hardware, networking, etc), a **container virtualises an entire operating system**. The significant difference between containers and VMs is that **containers share the host's operating system** with other containers.



Colour stories can be achieved by **using the same colour for text and graphics that represent the same concept**. It is often even more effective maintaining this consistency throughout an entire presentation.

Images and graphics

Quality and size: Use high-quality images and ensure that they are appropriately sized (with the correct aspect ratio) and do not pixelate when projected.

1) The main components of Amazon VPC

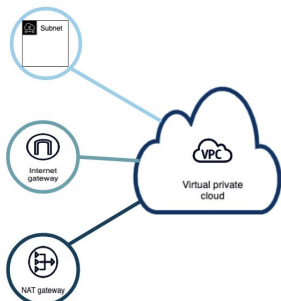
When setting up your Amazon VPC it is important to consider each of the following components and set it up accordingly. The first few components to consider: **Subnets**, an **Internet gateway** and a **NAT gateway**.

- Create separate **subnets** for different purposes.
- **Public** subnet - visible to the outside world - attach a public IP address.
- **Private** subnet - not publicly available - backend information you do not want visible.
- A subnet can only be defined in one **availability zone**.

- An **Internet gateway** allows access to and from the outside world.
- It does not have an IP address and is **managed by AWS**.
- If you need to access your *AWS resources from anywhere they have to be in a public subnet, with a **route** defined to the internet gateway.

- Create a **NAT gateway** for AWS resources that can't be visible, but requires access to the outside world.
- Select the subnet for the gateway and **AWS will manage** the rest.
- It **allows outbound** traffic but **limits inbound** access.

*Consider AWS resources to be a virtual machines/instances in this scenario.



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Networking and content delivery

The main components of Amazon VPC

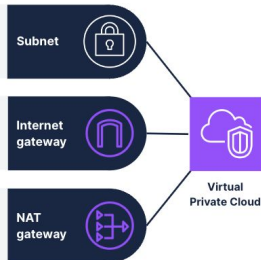
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We can **maintain the correct aspect ratio** of images and graphics by **holding the SHIFT key** when we resize them.

Images and graphics

Purpose: Images should enhance our message, not distract from it. These images or graphics should also be relevant to the message we are trying to convey.

Service Control Policies

Service Control Policies (SCPs) offer central control over the maximum available permissions for all accounts in your organisation.

Service control Policies are available only in an organisation that has all features enabled, including consolidated billing.

Service Control Policies are similar to IAM permission policies. However, they never grant permissions, instead, they specify the maximum permissions for an organisation

Attaching a **Service Control Policy** to an **Organisational Unit** defines a safeguard for the actions that accounts in that **OU** can actually do.

SCPs should not be used as a substitute for IAM configurations within each account.



AWS cloud security

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Attaching a Service Control Policy to an **Organisational Unit (OU)** defines a **safeguard** for the **actions that accounts in that OU can actually do**.



SCPs should **not be used as a substitute for IAM configurations** within each account.

Rather than using images to fill gaps on our slide decks, we can **increase the size of our font**, **add more whitespace**, **add a subtle background shape**, and/or **decrease the proximity** between text paragraphs.

Transitions and animations

Transitions and animations: Use slide transitions and animations sparingly and purposefully. They can enhance engagement but should not distract from your message.



Data storage fundamentals

Cloud computing service models

There are three main cloud computing service models, namely Software-as-a-Service (SaaS), Platform-as-a-Service (PaaS), and Infrastructure-as-a-Service (IaaS).

5

We naturally read from **top to bottom, left to right**, so it is important to time each animation after one another in a flow that does not break the way we consume the information.

Summary

In this train, we looked at the key principles of creating an effective presentation and how to implement them.

